WORKED SOLUTION

STEP II 2010, Question 7

Pure • Cubic Equations

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/10-s2-q7

The key idea

A cubic with positive leading coefficient crosses the axis once only exactly when its two turning points lie on the same side of the axis. Part (ii) is the classical trick for solving cubics with no x^2 term: the substitution $x = u + q/u$ is chosen so that the cross terms in x^3 cancel against $-3qx$, which leaves a quadratic in u^3. Part (iii) then turns out to be the same cubic in disguise, with p in place of x.

Solution

Part (i)

Write $f(x) = x^3 - 3qx - q(1+q)$. Then $f'(x) = 3x^2 - 3q$, which vanishes at $x = \pm\sqrt{q}$ (real, since $q > 0$). Since $f''(x) = 6x$, the point at $x = -\sqrt{q}$ is a local maximum and the point at $x = \sqrt{q}$ is a local minimum.

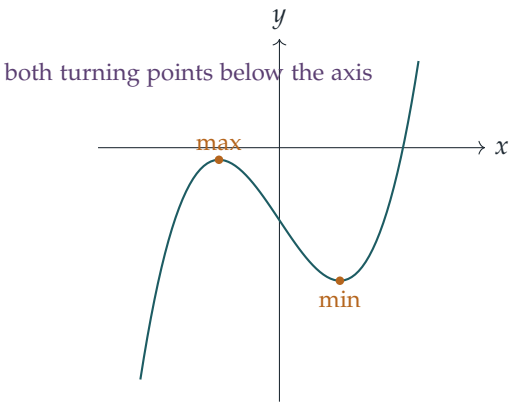
At the local maximum,

$$\begin{aligned} f(-\sqrt{q}) &= -q\sqrt{q} + 3q\sqrt{q} - q - q^2 \\ &= -q(q - 2\sqrt{q} + 1) \\ &= -q(\sqrt{q} - 1)^2. \end{aligned}$$

Since $q > 0$ and $q \neq 1$ (so $\sqrt{q} \neq 1$), this is strictly negative. The local minimum is lower still:

$$f(\sqrt{q}) = q\sqrt{q} - 3q\sqrt{q} - q - q^2 = -2q\sqrt{q} - q - q^2 < 0.$$

So both turning points lie below the x-axis. For $x < \sqrt{q}$ the curve never rises above its local maximum value, which is negative, so it does not meet the axis there. For $x > \sqrt{q}$ the function is strictly increasing, from a negative value to $+\infty$, so it crosses the axis exactly once. Hence the curve crosses the x-axis once only.



Part (ii)

Cubing $x = u + q/u$ with the binomial theorem,

$$\begin{aligned} x^3 &= u^3 + 3u^2 \cdot \frac{q}{u} + 3u \cdot \frac{q^2}{u^2} + \frac{q^3}{u^3} \\ &= u^3 + \frac{q^3}{u^3} + 3q\left(u + \frac{q}{u}\right) = u^3 + \frac{q^3}{u^3} + 3qx. \end{aligned}$$

So $x^3 - 3qx = u^3 + q^3/u^3$, and the cubic becomes

$$u^3 + \frac{q^3}{u^3} = q(1+q).$$

Multiplying by u^3 (note $u \neq 0$ for the substitution to make sense), the quadratic equation satisfied by u^3 is

$$(u^3)^2 - q(1+q)\,u^3 + q^3 = 0.$$

Ask yourself How does this quadratic factorise?

The two roots must have sum $q + q^2$ and product $q^3 = q \cdot q^2$, so they are q and q^2:

$$(u^3 - q)(u^3 - q^2) = 0.$$

Taking real cube roots, $u = q^{1/3}$ or $u = q^{2/3}$. Both give the same x:

$$q^{1/3} + \frac{q}{q^{1/3}} = q^{1/3} + q^{2/3}, \qquad q^{2/3} + \frac{q}{q^{2/3}} = q^{2/3} + q^{1/3}.$$

As a direct check, with $s = q^{1/3}$ and $x = s + s^2$ we have $x^3 = s^3 + 3s^4 + 3s^5 + s^6$ and $3qx = 3s^4 + 3s^5$, so $x^3 - 3qx = s^3 + s^6 = q(1+q)$. By part (i) the cubic has only one real root when $q > 0$, $q \neq 1$, so the real root is

$$x = q^{1/3} + q^{2/3}.$$

Part (iii)

Since $\alpha + \beta = p$ and $\alpha\beta = q$,

$$\alpha^3 + \beta^3 = (\alpha + \beta)^3 - 3\alpha\beta(\alpha + \beta) = p^3 - 3qp,$$

as required.

If one root is the square of the other, then either $\alpha^2 = \beta$ or $\beta^2 = \alpha$, so in either case

$$(\alpha^2 - \beta)(\beta^2 - \alpha) = 0.$$

This expression is symmetric in α and β, so we can write it in terms of p and q. Expanding,

$$\begin{aligned}(\alpha^2 - \beta)(\beta^2 - \alpha) &= \alpha^2\beta^2 - (\alpha^3 + \beta^3) + \alpha\beta \\ &= q^2 - (p^3 - 3qp) + q.\end{aligned}$$

Setting this to zero gives the relationship

$$p^3 - 3qp - q(1 + q) = 0.$$

This is exactly the cubic of part (ii) with p in place of x. When $q > 0$, $q \neq 1$ and p is real, p must be its only real root, so

$$p = q^{1/3} + q^{2/3}.$$

This makes sense: the roots are then $\alpha = q^{1/3}$ and $\beta = q^{2/3} = \alpha^2$, whose sum is p and product is q.

Checks

Take $q = 8$: part (ii) gives $x = 2 + 4 = 6$, and indeed $216 - 144 - 72 = 0$. In part (iii), $p = 6$ gives $t^2 - 6t + 8 = (t - 2)(t - 4)$, with $4 = 2^2$. A computer confirmed both turning-point values in (i) for thousands of random q, the identity $x^3 - 3qx = u^3 + q^3/u^3$ for random complex u, and that $p = q^{1/3} + q^{2/3}$ always yields a pair of roots with one the square of the other.

Common slips

- In (i), checking only that the local minimum is negative. That alone allows three crossings; it is the local maximum being negative that matters, and that is where $q \neq 1$ is used.
- In (ii), worrying that the two values of u^3 give two different roots. They give the same x, as they must, because u and q/u play symmetric roles.

Beyond STEP

The substitution in (ii) is the heart of Cardano's formula for the general cubic, and it is where complex numbers first became unavoidable in mathematics: when a cubic has three real roots, the quadratic for u^3 has complex roots. The excluded case $q = 1$ is a nice illustration. Then $p = -1$ is also allowed, and $t^2 + t + 1 = 0$ has roots ω and ω^2, the complex cube roots of unity, each the square of the other. The full story of when polynomial equations can be solved by radicals is Galois theory, told well in Allenby, *Rings, Fields and Groups*.

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